pyALDIC Quick Guide

Two-page reference. Full walkthrough: `user-guide.tex`.

1. Install & launch

Pick one install path (all are equivalent at runtime):

- **Windows, no Python**: download `pyALDIC-*-win64.zip` from the latest release, unzip it, double-click `pyALDIC.exe`. Nothing to install. SmartScreen will ask once: *More info → Run anyway.*
- **From PyPI** (easiest with Python): `pip install al-dic`.
- **From a GitHub Release wheel**: download `al_dic-*.whl` from github.com/zachtong/pyALDIC/releases and run `pip install al_dic-*.whl`. Useful when behind a firewall that blocks PyPI.
- **From source** (development): `git clone https://github.com/zachtong/pyALDIC.git` then `pip install -e ".[dev]"` inside the clone.

The Python paths require Python ≥ 3.10; launch with `al-dic` (or `python -m al_dic`). The Windows bundle launches from `pyALDIC.exe`.

The first analysis after installing compiles the solver and takes noticeably longer; pyALDIC starts that in the background at launch and caches it, so it happens once.

2. Load images

Drop an image folder onto the drop zone or click Browse. All files must be the same size. Frame 1 is always the reference. Frames stream from disk on demand, so hundreds of 4K images load without exhausting RAM.

3. Choose Workflow Type

This decides which frames need a Region of Interest, so do it *before* drawing.

Experiment	Tracking + Ref Update
Soft material <2% strain	Accumulative
2–30% strain, any rotation >5°, impact	Incremental + Every Frame
Known quasi-static load steps	Incremental + Custom Frames

Solver: **AL-DIC** default (smoother, ∼ 3× slower); **Local DIC** for quick iteration.

4. Draw Region of Interest

The blue-accent hint above the toolbar tells you which frames need a region. Toolbar:

- **+ Add**, **Cut**, **+ Refine** (brush, frame 1 only). Each opens a Polygon / Rectangle / Circle menu.
- **Import** / **Batch Import** – load PNG masks.
- **Save** / **Invert** / **Clear** – per- frame mask operations.

For per-frame editing, click the Need / Edit / Add button on a row in the image list.

5. Parameters

Control	Guidance
Subset Size	41 default; larger (51–81) for noise, smaller (21–31) for fine speckle.
Subset Step	8 for 512^2, 16 for 1024^2. Halving step → ∼ 4× runtime.
Search Range	20 default; raise for large inter-frame motion. Auto-expand is on.
Refinement	Check Inner / Outer to densify mesh near hole or ROI boundaries; pick a level (Light to Ultra).

Advanced section (collapsed by default) holds the initial-guess mode and auto-expand toggle; defaults are fine.

Starting Points (special init-guess mode). Select this radio for large displacement (> 50 px) or discontinuous fields (cracks, shear bands). Every connected mask region shows as yellow on the canvas; place at least one point per region (*Place Starting Points* to click, or *Auto-place* to pick the highest- NCC node per region) until every region turns green. Run stays disabled until that's true. On a reference-frame switch the points are warped automatically; if warping fails the pipeline auto-places fresh points on the new reference and continues.

6. Run & Cancel

Click the blue **Run DIC Analysis**. Progress, elapsed / remaining, and console log appear on the right. Red **Cancel** stops cleanly (partial results kept).

Fatal errors raise a modal dialog; the full traceback is in the console log.

7. View results

On the right sidebar after a run:

- **FIELD**: Disp U / Disp V toggle. *Show on deformed frame* checkbox sits here (controls WHERE, not how).
- **VISUALIZATION**: Colormap, Auto / Fixed range radio pair (Fixed = manual Min / Max), opacity. Numeric boxes accept both "." and "," decimals.
- **PHYSICAL UNITS**: Pixel size + frame rate – scales displacement / velocity output; strain is dimensionless.

8. Strain post-processing

A separate **Strain window** opens automatically when the run finishes.

- **Method**: *Plane fitting* (smooth, default) vs *FEM nodal* (higher resolution, noisier).
- **VSG size**: Virtual strain gauge diameter (default 41 px).
- **Smoothing**: Off / Light (0.5× step) / Medium (1× step) / Strong (2× step) ▲.
- **Strain type**: **Infinitesimal** (small deformation only), *Eulerian-Almansi*, **Green-Lagrangian** (**use for any rotation > 2°**; it is exactly zero under rigid rotation).

Click green **Compute Strain** whenever parameters change. Field selector now exposes `strain_exx`, `strain_eyy`, `strain_exy`, principal strains, `strain_maxshear`, `strain_von_mises`, `strain_rotation`, plus the displacement fields.

9. Export

Both windows have an **Export Results** button; they open the same 5-tab dialog:

- **Data**: NPZ / MAT / CSV of any selected fields.
- **Images**: per-field JPEG (quality 60–100) / PNG / TIFF per frame; resolution presets cap the long edge (512–2048 px or full). Colorbar ON by default.
- **Animation**: MP4 / GIF, encoded frame-by-frame (no RAM spike); *Frame step* keeps every Nth frame with playback duration preserved. Requires `ffmpeg` on PATH for MP4 (falls back to GIF otherwise).
- **Report**: multi-page PDF summary of the run.
- **Preview & Colorbar**: WYSIWYG preview through the real export path. Colorbar position / font / thickness / background / margin; field colormap–range–opacity synced two-way with Images; *Apply to all fields*.

10. Save / resume sessions

File → **Save Session** (`Ctrl+S`) writes a single `.aldic` bundle: every Region of Interest, all parameters, view state, and — optionally, a prompt shows the size estimate — the computed results. Async, with a progress dialog.

File → **Open Session** (`Ctrl+O`) restores the exact page (frame, field, colormap, ranges) with no recompute. Legacy `.aldic.json` files still open. If the image folder is gone, parameters and Regions of Interest still load; re-drop the images manually. Starting Points are not saved — re-place them before the next Run.

Windows: **File** → **Associate .aldic files** makes double-clicking a session file open it in pyALDIC.

11. Troubleshooting quick table

Symptom	Likely cause / fix
Run button disabled	No images or no Region of Interest on frame 1.
Crazy strain under big rotation	Using Infinitesimal strain; switch to Green-Lagrangian.
Strain map / export all blank or NaN	Plane-fit edge-trim removed every node — ROI too thin or VSG too large. Lower *Trim* (α), reduce VSG size, or use FEM nodal.
FFT peaks clipped warnings	Raise Search Range or keep auto-expand on.
Refine brush grayed out	Brush is frame-1 only; navigate to frame 1.
Results poor near hole edges	Enable *Refine Inner Boundary*.
Slow run	Halve Subset Step for next iteration; use Local DIC for previews.
Session won't load	Check console: corrupt file, future schema, or missing image folder (warning only – params restore).